

46 YEARS OF EXCELLENCE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-26** **Date: 25.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. With rise in temperature, the Young's modulus of elasticity
   A) changes erratically B) decreases
   C) increases D) remains unchanged

2. If R is the radius of the earth and the acceleration due to gravity on the surface of earth is $g = \pi^2 m/s^2$, then the length of the second's pendulum at a height $h = 2R$ from the surface of earth will be:

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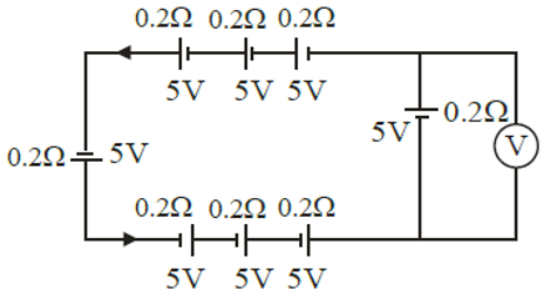
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A) $\frac{2}{9}m$ B) $\frac{1}{9}m$ C) $\frac{4}{9}m$ D) $\frac{8}{9}m$

3. A plane is in level flight at constant speed and each of its two wings has an area of $40m^2$. If the speed of the air is 180 km/h over the lower wing surface and $252\,km/h$ over the upper wing surface, the mass of the plane is _____ kg. (Take air density to be $1\,kg\ m^{-3}$ and $g = 10\ ms^{-2}$)

A) 10500 B) 9600 C) 12800 D) 8400

4. The reading in the ideal voltmeter (V) shown in the given circuit diagram is:



A) 5 V B) 10 V C) 0 V D) 3 V

5. Two identical capacitors have same capacitance C. One of them is charged to the potential V and other to the potential 2V. The negative ends of both are connected together. When the positive ends are also joined together, the decrease in energy of the combined system is:

A) $\frac{1}{4}CV^2$ B) $2CV^2$ C) $\frac{1}{2}CV^2$ D) $\frac{3}{4}CV^2$

6. Two moles a monoatomic gas is mixed with six moles of a diatomic gas. The molar specific heat of the mixture at constant volume is:

A) $\frac{9}{4}R$ B) $\frac{7}{4}R$ C) $\frac{3}{2}R$ D) $\frac{5}{2}R$

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7. A ball of mass 0.5 kg is attached to a string of length 50 cm. The ball is rotated on a horizontal circular path about its vertical axis. The maximum tension that the string can bear is 400 N. The maximum possible value of angular velocity of the ball in rad/s is:

A) 1600 B) 40 C) 1000 D) 20

8. A parallel plate capacitor ha a capacitance $C = 200\ pF$. It is connected to $230V$ ac supply with an angular frequency 300 rad/s. The rms value of conduction current in the circuit and displacement current in the capacitor respectively are

A) $1.38\ \mu A$ and $1.38\ \mu A$ B) $14.3\ \mu A$ and $143\ \mu A$

C) $13.8\ \mu A$ and $138\ \mu A$ D) $13.8\ \mu A$ and $13.8\ \mu A$

9. The pressure and volume of an ideal gas are related as $PV^{3/2} = K$ (Constant). The work done when the gas is taken from state $A(P_1, V_1, T_1)$ to state $B(P_2, V_2, T_2)$ is:

A) $2(P_1V_1 - P_2V_2)$ B) $2\left(P_2V_2 - \sqrt{P_1V_1}\right)$

C) $2\left(\sqrt{P_1}V_1 - \sqrt{P_2}V_2\right)$ D) $2\left(P_2\sqrt{V_2} - P_2\sqrt{V_1}\right)$

10. A galvanometer has a resistance of $50\ \Omega$ and it allows maximum current of $5\ mA$. It can be converted into voltmeter to measure upto $100V$ by connecting in series a resistor of resistance

A) $5975\ \Omega$ B) $20050\ \Omega$ C) $19950\ \Omega$ D) $19500\ \Omega$

11. The de Broglie wavelengths of a proton and an α particle are λ and 2λ respectively. The ratio of the velocities of proton and α particle will be:

A) 1:8 B) 1:2 C) 4:1 D) 8:1

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12. 10 divisions on the main scale of a Vernier calliper coincide with 11 divisions on the Vernier scale. If each division on the main scale is of 5 units, the least count of the instrument is:

A) $\frac{1}{2}$ B) $\frac{10}{11}$ C) $\frac{50}{11}$ D) $\frac{5}{11}$

13. In series LCR circuit, the capacitance is changed from C to 4C. To keep the resonance frequency unchanged, the new inductance should be:

A) reduced by $\frac{1}{4}L$ B) increased by $2L$

C) reduced by $\frac{3}{4}L$ D) increased to $4L$

14. The radius (r), length (l) and resistance (R) of a meatal wire was measured in the laboratory as

$r=(0.35\pm0.05)cm$

$R=(100\pm10)ohm$

$l=(15\pm0.2)cm$

The percentage error in resistivity of the material of the wire is:

A) 25.6% B) 39.9% C) 373% D) 35.6%

15. The dimensional formula of angular impulse is:

A) $\left[ML^{-2}T^{-1}\right]$ B) $\left[ML^{2}T^{-2}\right]$ C) $\left[MLT^{-1}\right]$ D) $\left[ML^{2}T^{-1}\right]$

16. A simple pendulum of length $1m$ has a wooden bob of mass 1 kg. It is struck by a bullet of mass $10^{-2}kg$ moving with a speed of $2\times10^{2}ms^{-1}$. The bullet gets embedded into the bob. The height to which the bob rises before swinging back is.

(use $g=10m/s^{2}$)

A) $0.30m$ B) $0.20m$ C) $0.35m$ D) $0.40m$

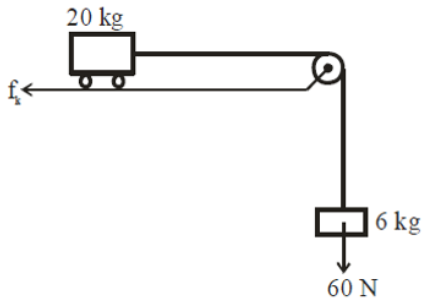
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17. A particle moving in a circle of radius R with uniform speed takes time T to complete one revolution. If this particle is projected with the same speed at the angle θ to the horizontal, the maximum height attained by it is equal to $4R$. The angle of projection θ is then given by:

A) $\sin^{-1}\left[\frac{2gT^2}{\pi^2 R}\right]^{\frac{1}{2}}$ B) $\sin^{-1}\left[\frac{\pi^2 R}{2gT^2}\right]^{\frac{1}{2}}$ C) $\cos^{-1}\left[\frac{2gT^2}{\pi^2 R}\right]^{\frac{1}{2}}$ D) $\cos^{-1}\left[\frac{\pi R}{2gT^2}\right]^{\frac{1}{2}}$

18. Consider a block and trolley system as shown in figure. If the coefficient of kinetic friction between the trolley and the surface is 0.04, the acceleration of the system in ms^{-2} is: (Consider that the string is massless and unstretchable and the pulley is also massless and frictionless):



A) 3 B) 4 C) 2 D) 1.2

19. The minimum energy required by a hydrogen atom in ground state to emit radiation in Balmer series is nearly:

A) 1.5 eV B) 13.6 eV C) 1.9 eV D) 12.1 eV

20. A monochromatic light of wavelength $6000\,\text{Å}$ is incident on the single slit of width $0.01mm$. If the diffraction pattern is formed at the focus of the convex lens of focal length $20cm$, the linear width of the central maximum is:

A) $60mm$ B) $24mm$ C) $120mm$ D) $12mm$

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A regular polygon of 6 sides is formed by bending a wire of length 4π meter. If an electric current of $4\pi\sqrt{3}\ A$ is flowing through the sides of the polygon, the magnetic field at the centre of the polygon would be $x\times10^{-7}T$. The value of x is

22. A rectangular loop of sides $12cm$ and $5cm$, with its sides parallel to the x-axis and $y-$axis respectively moves with a velocity of $5cm/s$ in the positive x axis direction, in a space containing a variable magnetic field in the positive z direction. The field has a gradient of $10^{-3}T/cm$ along the negative x direction and it is decreasing with time at the rate of $10^{-3}T/s$. If the resistance of the loop is $6\ m\Omega$, the power dissipated by the loop as heat is _____$\times10^{-9}W$.

23. The distance between object and its 3 times magnified virtual image as produced by a convex lens is $20\ cm$. The focal length of the lens used is _____cm.

24. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is $1.5\ g/cc$, the dielectric constant of water will be _____ (Take density of water = $1\ g/cc$)

25. The radius of a nucleus of mass number 64 is 4.8 fermi. Then the mass number of another nucleus having radius of 4 fermi is $\frac{1000}{x}$, where x is _____.

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. If one strand of a DNA has the sequence ATGCTTCA, sequence of the bases in complementary strand is:

A) CATTAGCT B) TACGAAGT

C) GTACTTAC D) ATGCGACT

27. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Haloalkanes react with KCN to form alkyl cyanides as a main product while with AgCN form isocyanide as the main product.

Reason (R) : KCN and AgCN both are highly ionic compounds.

In the light of the above statement, choose the most appropriate answer from the options given below:

A) (A) is correct but (R) is not correct

B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

C) (A) is not correct but (R) is correct

D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

28. In acidic medium, $K_2Cr_2O_7$ shows oxidizing action as represented in the half reaction.

$Cr_2O_7^{2-} + XH^+ + Ye^- \rightarrow 2A + ZH_2O$

X, Y, Z and A are respectively are:

A) 8, 6, 4 and Cr_2O_3 B) 14, 7, 6 and Cr^{3+}

C) 8, 4, 6 and Cr_2O_3 D) 14, 6, 7 and Cr^{3+}

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29. Which of the following reactions are disproportionation reactions?

(a) $Cu^{+} \rightarrow Cu^{2+} + Cu$

(b) $3MnO_4^{2-} + 4H^{+} \rightarrow 2MnO_4^{-} + MnO_2 + 2H_2O$

(c) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

(d) $2MnO_4^{-} + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^{+}$

Choose the correct answer from the options given below:

A) (a), (b) B) (b), (c), (d) C) (a), (b), (c) D) (a), (d)

30. In case of isoelectronic species the size of F^{-}, Ne and Na^{+} is affected by:

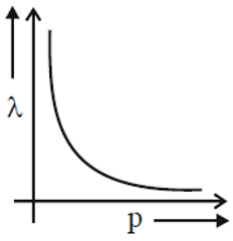
A) Principal quantum number (n)

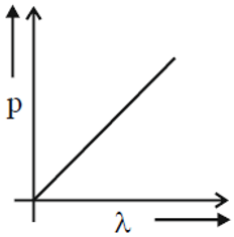
B) None of the factors because their size is the same

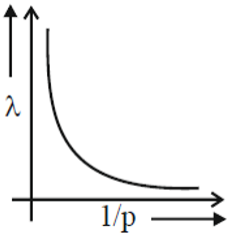
C) Electron – electron interaction in the outer orbitals

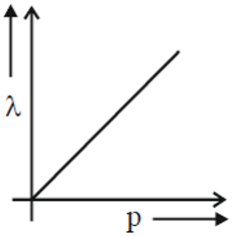
D) Nuclear charge (z)

31. According to the wave-particle duality of matter by de-Broglie, which of the following graph plot presents most appropriate relationship between wavelength of electron (λ) and momentum of electron (p)

A) 

B) 

C) 

D) 

32. Given below are two statements:

Statement (I): A solution of $[Ni(H_2O)_6]^{2+}$ is green in colour.

Statement (II): A solution of $[Ni(CN)_4]^{2-}$ is colourless.

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In the light of the above statements, choose the most appropriate answer from the options given below:

A) Both Statement I and Statement II are incorrect

B) Both Statement I and Statement II are correct

C) Statement I is incorrect but Statement II is correct

D) Statement I is correct but Statement II is incorrect

33. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : PH_3 has lower boiling point than NH_3.

Reason (R) : In liquid state NH_3 molecules are associated through vander waal's forces, but PH_3 molecules are associated through hydrogen bonding.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

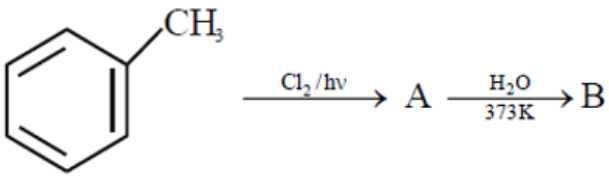
A) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

B) (A) is not correct but (R) is correct

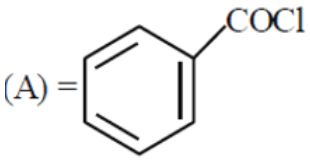
C) Both (A) and (R) are correct but (R) is the correct explanation of (A)

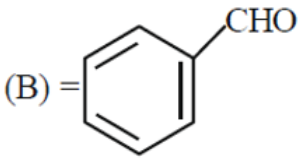
D) (A) is correct but (R) is not correct

34. Identify A and B in the following sequence of reaction

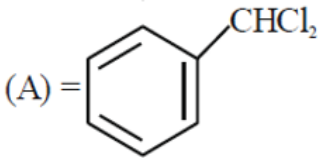


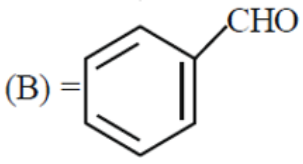
A)





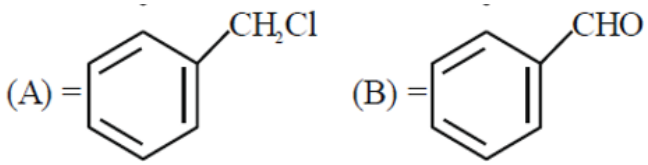
B)

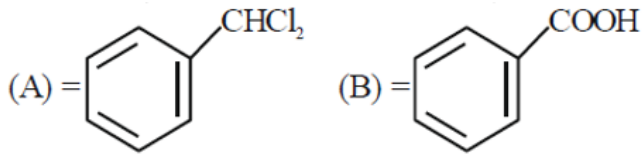




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C) 

D) 

35. Given below are two statements:

Statement (I) : Aminobenzene and aniline are same organic compounds.

Statement (II) : Aminobenzene and aniline are different organic compounds.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

A) Both Statement I and Statement II are correct

B) Statement I is correct but Statement II is incorrect

C) Statement I is incorrect but Statement II is correct

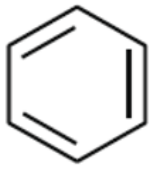
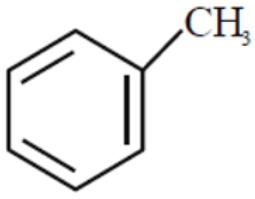
D) Both Statement I and Statement II are incorrect

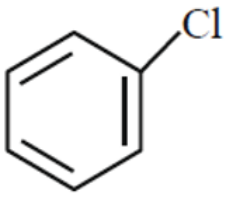
36. Which of the following complex is homoleptic?

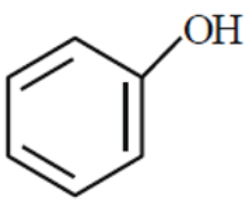
A) $\left[Ni(CN)_4\right]^{2-}$ B) $\left[Ni(NH_3)_2Cl_2\right]$

C) $\left[Fe(NH_3)_4Cl_2\right]^{+}$ D) $\left[Co(NH_3)_4Cl_2\right]^{+}$

37. Which of the following compound will most easily be attacked by an electrophile?

A)  B) 

C) 

D) 

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38. Ionic reactions with organic compounds proceed through:

(a) Homolytic bond cleavage

(b) Heterolytic bond cleavage

(c) Free radical formation

(d) Primary free radical

(e) Secondary free radical

Choose the correct answer from the options given below:

A) (a) only B) (c) only C) (b) only D) (d) and (e) only

39. Arrange the bonds in order of increasing ionic character in the molecules.

LiF, K_2O, N_2, SO_2 and ClF_3.

A) $ClF_3 < N_2 < SO_2 < K_2O < LiF$ B) $LiF < K_2O < ClF_3 < SO_2 < N_2$

C) $N_2 < SO_2 < ClF_3 < K_2O < LiF$ D) $N_2 < ClF_3 < SO_2 < K_2O < LiF$

40. We have three aqueous solutions of $NaCl$ labelled as 'A', 'B' and 'C' with concentration 0.1 M, 0.01 M and 0.001 M, respectively. The value of Van't hoff factor (i) for these solutions will be in the order.

A) $i_A < i_B < i_C$ B) $i_A < i_C < i_B$ C) $i_A = i_B = i_C$ D) $i_A > i_B > i_C$

41. In Kjeldahl's method for estimation of nitrogen, $CuSO_4$ acts as:

A) Reducing agent B) Catalytic agent

C) Hydrolysis agent D) Oxidising agent

42. Given below are two statements :

Statement (I) : Potassium hydrogen phthalate is a primary standard for standardisation of sodium hydroxide solution.

Statement (II) : In this titration phenolphthalein can be used as indicator.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

A) Both Statement I and Statement II are correct

B) Statement I is correct but Statement II is incorrect

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C) Statement I is incorrect but Statement II is correct

D) Both Statement I and Statement II are incorrect

43. Match List – I with List – II.

	List – I (Reactions)		**List –II (Reagents)**
a.	$CH_3(CH_2)_5-\underset{\underset{O}{\|\|}}{C}-OC_2H_5 \longrightarrow CH_3(CH_2)_5CHO$	I.	CH_3MgBr, H_2O
b.	$C_6H_5COC_6H_5 \longrightarrow C_6H_5CH_2C_6H_5$	II.	Zn(Hg) and conc.HCl
c.	$C_6H_5CHO \longrightarrow C_6H_5CH(OH)CH_3$	III.	$NaBH_4$, H^+
d.	$CH_3COCH_2COOC_2H_5 \longrightarrow CH_3\underset{\underset{H}{\|}}{C}(OH)CH_2COOC_2H_5$	IV.	DIBAL-H, H_2O

Choose the correct answer from options given below:

A) $a-III, b-IV, c-I, d-II$ B) $a-IV, b-II, c-I, d-III$

C) $a-IV, b-II, c-III, d-I$ D) $a-III, b-IV, c-II, d-I$

44. Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following:

A) $q=0, \Delta T \neq 0, w=0$ B) $q=0, \Delta T<0, w \neq 0$

C) $q \neq 0, \Delta T=0, w=0$ D) $q=0, \Delta T=0, w=0$

45. Given below are two statements:

Statement (I) : The NH_2 group in Aniline is ortho and para directing and a powerful activating group.

Statement (II) : Aniline does not undergo Friedel-Craft's reaction (alkylation and acylation).

In the light of the above statements, choose the most appropriate answer from the options given below :

A) Both Statement I and Statement II are correct

B) Both Statement I and Statement II are incorrect

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C) Statement I is incorrect but Statement II is correct

D) Statement I is correct but Statement II is incorrect

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Number of optical isomers possible for 2 – chlorobutane ……

47. The potential for the given half cell at 298 K is $(-)$…….. $\times 10^{-2} V$.

$2H^{+}_{(aq)} + 2e^{-} \rightarrow H_{2(g)}$

$\left[H^{+}\right] = 1M, P_{H_2} = 2atm$

(Given: $2.303\, RT / F = 0.06V$, $\log 2 = 0.3$)

48. The number of white coloured salts among the following is _____

a) $SrSO_4$ b) $Mg(NH_4)PO_4$ c) $BaCrO_4$ d) $Mn(OH)_2$

e) $PbSO_4$ f) $PbCrO_4$ g) $AgBr$ h) PbI_2

i) CaC_2O_4 j) $\left[Fe(OH)_2(CH_3COO)\right]$

49. Among the following oxide of p – block elements, number of oxides having amphoteric nature is

$Cl_2O_7, CO, PbO_2, N_2O, NO, Al_2O_3, SiO_2, N_2O_5, SnO_2$

50. The number of molecules/ion/s having trigonal bipyramidal shape is _____

$PF_5, BrF_5, PCl_5, \left[PtCl_4\right]^{2-}, BF_3, Fe(CO)_5$

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. A bag contains 8 balls, whose colours are either white or black. 4 balls are drawn at random without replacement and it was found that 2 balls are white and other 2 balls are black. The probability that the bag contains equal number of white and black balls is:

A) $\frac{2}{5}$ B) $\frac{2}{7}$ C) $\frac{1}{7}$ D) $\frac{1}{5}$

52. The value of the integral $\int_0^{\frac{\pi}{4}} \frac{xdx}{\sin^4(2x)+\cos^4(2x)}$ equals:

A) $\frac{\sqrt{2}\pi^2}{8}$ B) $\frac{\sqrt{2}\pi^2}{16}$ C) $\frac{\sqrt{2}\pi^2}{32}$ D) $\frac{\sqrt{2}\pi^2}{64}$

53. If $A=\begin{bmatrix}\sqrt{2} & 1\\ -1 & \sqrt{2}\end{bmatrix}, B=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}, C=ABA^T$ and $X=A^TC^2A$, then $\det X$ is equal to:

A) 243 B) 729 C) 27 D) 891

54. If $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}, \tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and

$\tan C=\left(x^{-3}+x^{-2}+x^{-1}\right)^{\frac{1}{2}}, 0<A,B,C<\frac{\pi}{2}$, then $A+B$ is equal to:

A) C B) $\pi-C$ C) $2\pi-C$ D) $\frac{\pi}{2}-C$

55. If n is the number of ways five different employees can sit into four indistinguishable offices where any office may have any number of persons including zero, then n is equal to:

A) 47 B) 53 C) 51 D) 43

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56. Let $S=\left\{z\in C:|z-1|=1 \text{ and } \left(\sqrt{2}-1\right)\left(z+\bar{z}\right)-i\left(z-\bar{z}\right)=2\sqrt{2}\right\}$. Let $z_1, z_2 \in S$ be such that $|z_1|=\max\limits_{z\in S}|z|$ and $|z_2|=\min\limits_{z\in S}|z|$. Then $\left|\sqrt{2}z_1-z_2\right|^2$ equals:

A) 1 B) 4 C) 3 D) 2

57. Let the median and the mean deviation about the median of 7 observation 170, 125, 230, 190, 210, a, b be 170 and $\frac{205}{7}$ respectively. Then the mean deviation about the mean of these 7 observations is:

A) 31 B) 28 C) 30 D) 32

58. Let $\vec{a}=-5\hat{i}+\hat{j}-3\hat{k}, \vec{b}=\hat{i}+2\hat{j}-4\hat{k}$ and $\vec{c}=\left(\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right)\times\hat{i}\right)\times\hat{i}$. Then $\vec{c}.\left(-\hat{i}+\hat{j}+\hat{k}\right)$ is equal to

A) -12 B) -10 C) -13 D) -15

59. Let $S=\left\{x\in R:\left(\sqrt{3}+\sqrt{2}\right)^x+\left(\sqrt{3}-\sqrt{2}\right)^x=10\right\}$. Then the number of elements in S is:

A) 4 B) 0 C) 2 D) 1

60. The area enclosed by the curves $xy+4y=16$ and $x+y=6$ is equal to:

A) $28-30\log_e 2$ B) $30-28\log_e 2$

C) $30-32\log_e 2$ D) $32-30\log_e 2$

61. Let $f:R\to R$ and $g:R\to R$ be defined as

$f(x)=\begin{cases}\log_e x, & x>0\\ e^{-x}, & x\le 0\end{cases}$ and $g(x)=\begin{cases}x, & x\ge 0\\ e^x, & x<0\end{cases}$. Then, $gof:R\to R$ is

A) one-one but not onto B) neither one-one nor onto

C) onto but bot one-one D) both one-one and onto

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62. If the system of equations

$2x+3y-z=5$

$x+\alpha y+3z=-4$

$3x-y+\beta z=7$ has infinitely many solutions, then $13\alpha\beta$ is equal to

A) 1110 B) 1120 C) 1210 D) 1220

63. For $0<\theta<\pi/2$, if the eccentricity of the hyperbola $x^2-y^2\cos ec^2\theta=5$ is $\sqrt{7}$ times eccentricity of ellipse $x^2\cos ec^2\theta+y^2=5$, then the value of θ is:

A) $\frac{\pi}{6}$ B) $\frac{5\pi}{12}$ C) $\frac{\pi}{3}$ D) $\frac{\pi}{4}$

64. Let $y=y(x)$ be the solution of the differential equation

$\frac{dy}{dx}=2x(x+y)^3-x(x+y)-1,\ y(0)=1.$ Then, $\left(\frac{1}{\sqrt{2}}+y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ equals:

A) $\frac{4}{4+\sqrt{e}}$ B) $\frac{3}{3-\sqrt{e}}$ C) $\frac{2}{1+\sqrt{e}}$ D) $\frac{1}{2-\sqrt{e}}$

65. Let $f:R\to R$ be defined as

$$f(x)=\begin{cases}\frac{a-b\cos 2x}{x^2} & ; \quad x<0\\ x^2+cx+2 & ; \quad 0\le x\le 1\\ 2x+1 & ; \quad x>1\end{cases}$$

If f is continuous everywhere in R and m is the number of points where f is NOT differentiable then $m+a+b+c$ equals:

A) 1 B) 4 C) 3 D) 2

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66. Let $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$ be an ellipse, whose eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is $\sqrt{14}$. Then the square of the eccentricity of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

A) 3 B) 7/2 C) 3/2 D) 5/2

67. Let $3,a,b,c$ be in A.P. and $3,a-1,b+1,c+9$ be in G.P. Then, the arithmetic mean of a,b and c is:

A) –4 B) –1 C) 13 D) 11

68. Let $C: x^2+y^2=4$ and $C': x^2+y^2-4\lambda x+9=0$ be two circles. If the set of all values of λ so that the circles C and C' intersect at two distinct points, is $R-[a,b]$, then the point $(8a+12, 16b-20)$ lies on the curve:

A) $x^2+2y^2-5x+6y=3$ B) $5x^2-y=-11$

C) $x^2-4y^2=7$ D) $6x^2+y^2=42$

69. If $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2, \forall x\neq 0$ and $y=9x^2f(x)$, then y is strictly increasing in:

A) $\left(0,\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$ B) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

C) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$ D) $\left(-\infty,\frac{1}{\sqrt{5}}\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$

70. If the shortest distance between the lines $\frac{x-\lambda}{-2}=\frac{y-2}{1}=\frac{z-1}{1}$ and $\frac{x-\sqrt{3}}{1}=\frac{y-1}{-2}=\frac{z-2}{1}$ is 1, then the sum of all possible values of λ is:

A) 0 B) $2\sqrt{3}$ C) $3\sqrt{3}$ D) $-2\sqrt{3}$

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. If $x = x(t)$ is the solution of the differential equation $(t+1)dx = \left(2x + (t+1)^4\right)dt,\ x(0) = 2,$ then $x(1)$ equals _____.

72. The number of elements in the set $S = \{(x, y, z) : x, y, z \in Z,\ x + 2y + 3z = 42,\ x, y, z \geq 0\}$ equals _____ .

73. If the Coefficient of x^{30} in the expansion of $\left(1 + \frac{1}{x}\right)^6 \left(1 + x^2\right)^7 \left(1 - x^3\right)^8 ; x \neq 0$ is α, then $|\alpha|$ equals _____.

74. Let the line $L : \sqrt{2}x + y = \alpha$ pass through the point of the intersection P (in the first quadrant) of the circle $x^2 + y^2 = 3$ and the parabola $x^2 = 2y$. Let the line L touch two circles C_1 and C_2 of equal radius $2\sqrt{3}$. If the centres Q_1 and Q_2 of the circles C_1 and C_2 lie one the $y-$axis, then the square of the area of the triangle PQ_1Q_2 is equal to _____.

75. Let $A = \{1, 2, 3, ...20\}$. Let R_1 and R_2 two relation on A such that
$R_1 = \{(a, b) : b \text{ is divisible by a}\}$
$R_2 = \{(a, b) : a \text{ is an integral multiple of b}\}$.
Then, number of elements in $R_1 - R_2$ is equal to _____

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